

Parth Raju Gujarathi

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EDUCATION

- **University of California San Diego** Sept'25-June'27
M.S. in Computer Science GPA: 4.0
- **Sardar Patel Institute of Technology** Sept'20- July'24
B.Tech in Computer Engineering (Gold Medalist - Awarded for academic excellence) GPA: 4.0

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML, C

Developer Tools: Hugging Face, LangChain, CSS, ReactJS, NodeJS, Bootstrap, NextJS

Frameworks: PyTorch, TensorFlow, Scikit-learn, ExpressJS, Socket IO, Spring Boot, Flask

Cloud/Databases: Vector Databases, OracleSQL, MySQL, MongoDB, SAP Hana

Relevant Coursework: Recommender Systems & Data Mining, AI : Probabilistic Methods, AI Agents, Statistical NLP

WORK EXPERIENCE

- **Deutsche Bank AG - Software Engineer** July 2024 - September 2025
 - Contributed to the development of the Compliant Analytics System to strengthen cross-country trade compliance checks and streamline regulatory validation processes.
 - Implemented APIs to automate trade validations, automatically flagging transactions involving sanctioned countries or restricted goods, significantly reducing manual review time and enabling faster risk assessments.
- **LTIMindtree - Software Intern** January 2024 - June 2024
 - Developed an RFP (Request for Proposal) automation system using Retrieval-Augmented Generation, LangChain, and vector databases to improve document retrieval and automatically generate new proposals by synthesizing information from existing proposals and RFPs. Built Python-based Flask services to support backend processing and optimize system efficiency.
 - Developed an AI-driven candidate screening module using OpenAI and Hugging Face for resume parsing, skill extraction, and candidate-job matching. Implemented Node.js worker-thread based backend services and optimized MongoDB queries to support efficient candidate evaluation.

PROJECT EXPERIENCE

- **BanditMafia - Multi Agent Reasoning for Social Deduction** January 2026 - March 2026
A Mafia simulation where LLM agents infer hidden roles & improve strategy via contextual bandit.
 - Engineered a role aware multi-agent LLM pipeline for social deduction, combining structured state extraction, belief map updates and evidence driven voting; improved villager win rate from 10% baseline to 35% (3.5×).
 - Implemented contextual Thompson Sampling for intent selection with round level telemetry (state, action, reward, outcome), enabling policy learning across self play and measurable strategy improvement over static prompting
- **Neurosafe - Advancing Mental Health using AI, ML and Blockchain** August 2023 - January 2024
An AI platform providing mental health support and secure record-keeping of user data.
 - Tools & technologies used: Python, NextJS, Flask, Scikit-learn, TensorFlow, Keras, IPFS
 - Developed a platform that analyzes user emotions using BERT (96% accuracy) and incorporates blockchain for secure record storage. Integrated a generative AI chatbot using Seq2Seq with LSTM for real-time support.
 - Published a paper at IEEE 2024 3rd International Conference for Innovation in Technology (INOCON).
- **Psychological Distress Detection and Classification** January 2023 - May 2023
A machine learning project using NLP to detect stress in users based on their social media posts.
 - Tools & technologies used: Python, NextJS, Flask, Scikit-learn, TensorFlow, Keras.
 - Developed a model using various algorithms to detect stress, achieving an accuracy of 98%. Employed a questionnaire-based survey for severity prediction with a 99.4% accuracy.
 - Published the findings at the IEEE 2024 3rd International Conference for Innovation in Technology (INOCON).
- **OneHub** February 2022 - May 2022
A website providing a one-stop solution for scheduling exams, assignments, and events for students.
 - Tools & technologies used: VueJS, NodeJS, ExpressJS, MongoDB.
 - A unified college portal enabling students to manage their tasks on one platform, featuring a schedule manager, announcement page, event page, and library page.

EXTRACURRICULAR ACTIVITIES

- Planned and executed various events as the Vice Chairperson of F.A.C.E, the technical and cultural committee of Computer Department of S.P.I.T such as coding events and sports competitions (2022-2023)
- Volunteered at the Abhyudaya program by Sardar Patel Jain Institute of Management and Research; assisted in teaching classes and taught elementary subjects to underprivileged primary students under this program (2023)
- Participated in various hackathons including JPMC's Code for Good for 2 consecutive years; collaborated with fellow developers to develop effective solutions for real-world problems (2022-2023)